INTERNAL · ADVISORY · JON ONLY

Oakwood *AI Innovation Lab*

Meeting Prep & Research

Companion document to the public briefing deck.
Everything you need to land the room. None of it appears in the deck.

PREPARED FOR

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MEETING

Dr. Shushannah Smith · Chair, Math & CS · Oakwood University

18 MAY 2026

Huntsville · Alabama

00 ORIENT

How to use *this document.*

This is the companion to the public deck at **https://oakwood-ai-hub.vercel.app/ai-lab-2026/**. The deck is what Dr. Smith sees. This is what *you* see — research, pricing detail, meeting strategy, objection handling, and risks. Nothing in here is on the deck. Nothing on the deck contradicts what is here.

Skim §3 (Hardware Detail) for the spec-recall you may need on the spot. Read §5 (Meeting Flow) once before walking in. Keep §6 (Objections) open mentally — Dr. Smith is a chair, she'll test the substance.

CONTEXT

01 What we know going in.

Confirmed references

Obsidian note · May 13, 2026 — "Oakwood University — Ms. Smith & AI Lab". Records that Dr. Smith runs Computer Science, wants the computer labs upgraded, wants to build an AI lab, has funding already identified, and is asking for guidance on what will actually be useful.

Priority Map · May 13, 2026. Action items: reach out to Dr. Smith, share the OU Campus Companion as a proof-of-delivery anchor, and prepare a recommendation memo covering lab setup, model/tool stack, governance, training, and student outcomes.

Draft Guard · today · 10:56 AM. Captured a long Oakwood AI Lab draft. This is the formalised version of that draft.

Not yet confirmed

There is no record of the outreach actually being sent, nor a full prior-meeting note. Treat this as a **warm first structured conversation**, not a follow-up. Open with a recognition note — not an assumption that prior context exists.

Live deliverables already in place

· Public deck deployed: `https://oakwood-ai-hub.vercel.app/ai-lab-2026/`
· GitHub repo: `github.com/Navigata1/oakwood-ai-hub`
· PDF advisory (this document): companion, not for sharing.

Tone guidance for the room

Empowering. Collaborative. Urgent-but-positive. Lead with student potential and Oakwood's leadership position as an HBCU. Avoid the salesman cadence. Translate every technical claim into a student outcome before moving on. Reference young builders as aspiration, never as pressure.

PERSON

02 Dr. Shushannah *Smith.*

Chair, Department of Mathematics & Computer Science, Oakwood University. Public record consistently describes her as deeply engaged in expanding minority participation in STEM and active in curriculum design. Student feedback skews to *encouraging and rigorous*. She is the decision-maker for this lab.

Role	Chair, Mathematics & Computer Science · Oakwood University
Mission lens	Advancing minority participation in STEM · HBCU excellence
Stated need	Upgrade computer labs · build an AI lab · already has funding
Asks for	Guidance on what is actually useful — not catalog dumps
Style	Professional, encouraging, expects rigor
Contact	ssmith@oakwood.edu (public record · verify card on arrival)
Decision power	Department-level approval on lab spend · institutional sign-off on larger packages

What she will likely value most

· **Student outcomes** she can name in conversation — internships, portfolios, demo days.
· **HBCU-specific framing** — Oakwood as a leader, not a catch-up.
· **Practicality** — vendor names, education pricing paths, faculty time budget.
· **Sustainability** — who maintains it, who teaches it, what year two looks like.
· **Governance** — institutional risk, FERPA-adjacent concerns, IP, audit trails.

HARDWARE

03 What's actually *available in 2026.*

All pricing is verified street/MSRP as of mid-May 2026. Always pull fresh quotes before procurement — memory prices in particular have moved twice this year. NVIDIA Academic, Apple Education, and CDW-G education pricing should be quoted alongside any direct price.

NVIDIA DGX Spark · GB10 Grace Blackwell Superchip

Chip	GB10 Grace Blackwell Superchip (NVIDIA + MediaTek)
CPU	20-core Arm — 10× Cortex-X925 + 10× Cortex-A725
GPU	Blackwell · 6,144 CUDA cores · 192 5th-gen Tensor cores · 4th-gen RT cores
Memory	128 GB unified LPDDR5X · 256-bit bus · ~273 GB/s
AI compute	1 PFLOP FP4 (sparsity)
Model ceiling	200B params (4-bit) on a single node
Pair-linked	256 GB / 405B params via ConnectX-7 between two units
Networking	200 Gbps
Storage	4 TB NVMe (Founder's Ed.) · 1 TB on most OEM partners
Form factor	150 × 150 mm · 1.2 kg · ~170 W
Price	$4,699 USD (MSRP since Feb 2026 — was $3,999 at launch; raised on memory supply)
OEM partners	ASUS Ascent GX10 · Dell Pro Max GB10 · MSI EdgeXpert MS-C931 · Acer Veriton GN100
Software	Pre-loaded NVIDIA AI stack · CUDA · NVIDIA OpenShell agent framework

Lab note. Arm-based — CUDA stack is excellent, but anything that requires an x86-only binary will need adaptation. Most modern AI tooling (PyTorch, vLLM, Ollama, llama.cpp, Hugging Face) ships Arm wheels. Power is moderate (~170 W under load). Heat output reasonable in a normal classroom HVAC.

Framework Desktop · AMD Ryzen AI Max+ 395 (Strix Halo)

CPU	16-core Zen 5 (32 threads) · up to 5.1 GHz · 55W TDP
iGPU	Radeon 8060S · 40 RDNA 3.5 Compute Units
Memory	128 GB LPDDR5X soldered · ~96 GB usable as VRAM
Storage	Dual M.2 2280 PCIe 4.0 NVMe slots (up to 16 TB each)
Network	5 GbE · Wi-Fi 7 · USB4 ports · 2× DisplayPort 2.1 · HDMI
Form factor	4.5 L mini-ITX · 400 W internal FlexATX PSU
Price · 128 GB	$1,999 USD launch · $2,566–$2,851 current with storage
Alternatives	Beelink GTR9 Pro ($2,999) · GEEKOM A9 Mega ($3,199) · HP Z2 Mini G1a · ASUS ROG Flow Z13
Why it matters	x86 — works with any standard AI tooling and Windows-first software

Lab note. AMD's own December 2025 benchmark put the Framework ~$/token *ahead of* DGX Spark on GPT-OSS 120B, GLM 4.5 Air, DeepSeek R1 Distill 70B. For pure student daily-driver + capable local AI, this is the most defensible economics in a 10-seat setup right now.

NVIDIA Jetson AGX Thor · Edge / Physical AI

Chip	Thor T5000 (Blackwell-class)
Memory	128 GB · 273 GB/s bandwidth
AI compute	Up to 2,070 FP4 TFLOPS
Use cases	Robotics · vision sensors · low-latency inference · humanoid prototypes
Developer kit	$3,499 USD
Why include	Gives the lab a "physical AI" hook that photographs well and recruits hardware-curious students

Apple Mac Studio · M4 Max / M3 Ultra-class

Unified memory up to 512 GB. Strong for media, vision, voice pipelines. Apple Core ML and MLX. Excellent for content / studio work, podcast and demo recording. Less ideal as the local-LLM workhorse for the *CUDA-default* tooling ecosystem, but valuable for creative pipelines and as a premium-tier seat in Research Studio / Flagship configurations. Education pricing typically 8–12% off list.

Networking, storage, power

· **Network:** 10 GbE minimum on the rack switch and uplinks; 25 GbE+ for $100K/$200K tiers.
· **NAS:** Synology RS-class or TrueNAS — snapshots, RBAC, dataset versioning.
· **Power:** Plan 30–40 A circuits on the lab side. DGX Spark is unexpectedly efficient (~170 W); the math hurts when you add multiple high-end PC workstations.
· **Cooling:** Room HVAC review is required at $100K+ tier. Lab can hit 4–5 kW peak.
· **MDM/Backup:** Jamf (if Apple-heavy), Intune (if Windows), Veeam or Rubrik for backup.

BUDGETS

04 What each tier *really costs.*

Numbers are calibrated to mid-May 2026 street pricing and exclude a ~10–15% buffer for peripherals, AV cabling, furniture (if needed), shipping, and contingency. **Always pull fresh quotes** — the right answer in the room is often “let me come back with a hard line-item by end of week.”

Starter Pod · $20,000

Pilot-focused. Quick wins. Proves the thesis inside one semester.

ITEM	QTY	UNIT $	EXT $
Mid-range 32–64 GB workstation/laptop	10	$1,100	$11,000
NVIDIA DGX Spark (1 TB OEM variant)	1	$4,300	$4,300
10 GbE managed switch (24-port)	1	$700	$700
NAS (Synology RS-class 4-bay + drives)	1	$1,400	$1,400
Demo display 65" + mount + adapters	1	$1,000	$1,000
Webcam + USB mic + ring light (capture)	1	$400	$400
Cables, peripherals, KVM	lot	—	$700

Subtotal hardware: $19,500. Add ~10–15% for cabling, AV, install, contingency.

Applied Builder · Recommended starting point · $50,000

Capable daily seats + real local AI experience. Production-grade demo days.

ITEM	QTY	UNIT $	EXT $
Framework Desktop AI Max+ 395 · 128 GB	10	$2,800	$28,000
NVIDIA DGX Spark (Founder's Edition)	2	$4,700	$9,400
25 GbE switch (uplinks) + 10 GbE access	1	$2,200	$2,200
NAS (Synology RS-class 8-bay + drives)	1	$3,400	$3,400
Demo wall 85" 4K + interactive mount	1	$2,500	$2,500
Studio capture (camera+mic+lighting kit)	1	$1,200	$1,200
Jetson Orin Nano Super (edge starter)	2	$500	$1,000
Instructor station + KVM + accessories	lot	—	$1,800

Subtotal hardware: $49,500. Add ~10–15% for cabling, AV, install, contingency.

Research Studio · $100,000

Capstones, fine-tuning, faculty research. Grant-credible from launch day.

ITEM	QTY	UNIT $	EXT $
Mac Studio M4 Max · 64 GB (premium seat)	10	$3,500	$35,000
NVIDIA DGX Spark (Founder's Edition)	3	$4,700	$14,100
RTX 5080-class GPU server (shared)	1	$8,000	$8,000
25 GbE backbone + 10 GbE access switching	1	$5,500	$5,500
Enterprise NAS (12-bay + drives + cache)	1	$8,500	$8,500
Demo wall 98" + secondary monitor pair	1	$5,500	$5,500
Studio rig (3-cam, lavalier, lighting)	1	$4,500	$4,500
Jetson AGX Thor Developer Kit	2	$3,500	$7,000
Robotics kits + vision sensors	lot	—	$4,500
MDM/Backup license bundle (Yr 1)	1	$3,200	$3,200

Subtotal hardware: $95,800. Add ~10–15% for cabling, AV, install, contingency.

Flagship Center · $200,000

Signature lab — recruitment piece, grant magnet, partnership stage.

ITEM	QTY	UNIT $	EXT $
Top-tier workstation (Mac Studio Ultra / RTX PRO)	10	$6,500	$65,000
NVIDIA DGX Spark cluster (4× w/ ConnectX-7)	4	$5,200	$20,800
Dedicated GPU server (4× H100 or B100-class)	1	$45,000	$45,000
100 GbE backbone + 25 GbE access	1	$14,000	$14,000
Enterprise NAS (24-bay all-flash tier)	1	$18,000	$18,000
Full studio AV (PTZ cams, console, lighting grid)	1	$15,000	$15,000
Jetson AGX Thor + humanoid/robotics fleet	lot	—	$12,000
Security appliance + audit logging stack	1	$6,500	$6,500
MDM/Backup/SSO licensing bundle	1	$4,500	$4,500

Subtotal hardware: $200,800. Add ~10–15% for cabling, AV, install, contingency.

Adjacent paths to keep in your back pocket

· **Hybrid Apple + PC.** Two-tier seating (5 Mac Studio for creative/voice/media work, 5 Framework for CUDA-default local AI). Diversifies the lab and matches student preference patterns. Often defensible at the $50K and $100K tiers.

· **Cloud-burst hybrid.** Lab handles privacy-bounded local work; AWS, GCP, or Azure education credits handle occasional fine-tuning or batch jobs that exceed local memory. Worth a $5–10K Yr-1 cloud credit allocation.

· **Refurbished enterprise.** Dell PowerEdge w/ A100 80GB on the used market runs $6–9K. Validate warranty path — Oakwood facilities may have a hard line on used gear.

· **Education pricing levers.** NVIDIA Academic Hardware program (DGX education discount, Jetson kits at deeper discount in bulk). Apple Education Store. CDW-G and Connection have HBCU-specific lines. NSF EAGER and HBCU UP programs occasionally subsidize purchase.

· **Modular AI cart.** One mobile rack with DGX + display + sensors that travels between rooms. Useful for outreach and for the rare class where the lab is double-booked.

MEETING

05 Flow, *questions, close.*

Primary objective

Build rapport. Understand exact funding, room, and timeline. Get agreement on a tier or a next diagnostic step. Leave with permission to send a procurement memo, room plan, and pilot curriculum on a specific date.

Opening — warm, recognizing, not presumptuous

> *"Dr. Smith, thank you for the time. I understand the department's been thinking about upgrading the labs with an AI focus, and I've come with a starting frame — but mostly I want to listen first. What does success look like for you on this?"*

The five questions you must walk out with answers to

01 Available funding range, source, and use-by deadline.

02 Room dimensions, current power/network capacity, and any constraint.

03 Preferred format — for-credit course, certificate, club, summer bridge, or faculty development.

04 Faculty capacity for oversight and mentoring · who would teach Year 1.

05 How they will *measure* the lab's success (portfolios, internships, grants, recruitment, retention).

Proof points to deploy naturally (not pitched)

· **OU Campus Companion** — when she asks about delivery speed or what you've shipped.
· **Sabbath Echo** — when faith-tech or multilingual capability comes up; demonstrates multi-language voice pipelines and Cloudflare/Supabase fluency.
· **YPE pipeline v6.5** — only if she asks how you'd run a curriculum at production cadence. Don't lead with it; it's an internal framework name.
· **Automotive shop** — if rapport calls for it. Real-world signal. Don't push.

Closing — concrete and timed

> *“If the $50K path resonates, I can have a line-item procurement memo and an 8-week pilot curriculum back to you by* ***Friday****. That gives the department something to react to before the next funding decision. Sound right?”*

06 OBJECTIONS

Eight pushbacks, *prepared answers.*

Q. “We already have computer labs.”

A. Agree visibly. “This isn't another computer lab — it's a production studio. The difference shows up in what walks out: a portfolio of shipped student work, not just completed coursework. The hardware reflects that — local AI nodes, demo wall, capture rig.”

Q. “Is all this local AI hardware really necessary? Can't we just use the cloud?”

A. “Cloud is the right answer for some of the work. Local matters for three things students need to learn: privacy-bounded data handling, predictable inference economics, and the actual engineering of running a model. Internships are asking for that fluency by name in 2026.”

Q. “Who maintains this once you're gone?”

A. “Two answers. One — the stack is intentionally boring under the hood (standard OS, standard tools, vendor support contracts). Two — by month three, students are running the lab. Peer mentoring is built into the curriculum on purpose. I'd also love to keep an advisory cadence with whoever leads it.”

Q. “Faculty don't have time to learn this.”

A. “The curriculum is designed so faculty don't need to be the AI experts on day one. Module 1 is the fluency layer for everyone, including faculty. I'd recommend running a one-week faculty bridge before the student cohort starts. After that, faculty become guides, not lecturers.”

Q. “Why not Apple? We have an Apple ecosystem.”

A. “Apple's a strong fit for media, voice, and creative pipelines. The $100K tier I prepared puts Mac Studio at every seat for that reason. Where we want NVIDIA is the shared local-LLM bench — it's where the CUDA tooling lives. A hybrid lab is actually the most realistic 2026 config.”

Q. “What about FERPA and student data?”

A. “Exactly why local matters. Module 5 makes data classification and audit logging a daily habit. We'd draft a lab use policy together — I can have a template ready by Friday.”

Q. “How do we measure if this is working?”

A. “Four signals I'd watch in semester one: number of shipped student artifacts, demo-day attendance, internship placements that cite the lab on a resume, and faculty self-reported AI confidence. I'll send a metrics dashboard proposal as part of the follow-on.”

Q. “Can we start smaller?”

A. “Yes — that's the $20K Starter Pod. It proves the thesis in one semester and leverages into a stronger grant case for the $50K or $100K expansion. The first demo day is the funding event.”

RISKS

07 What can go wrong, *and how to soften it.*

Procurement timing	University purchasing cycles can run 60–120 days. Always start with the pilot config that can ship inside one semester. Pre-stage vendor relationships now.
Memory price volatility	LPDDR5X / HBM pricing has moved twice in 2026. Quote validity is often 30 days. Lock orders when funding clears.
Arm vs x86 fragmentation	DGX Spark is Arm. 95% of AI tooling is fine; 5% (legacy Windows-only tools, some CAD) is not. Frame in the deck as "shared local node" and keep student seats on x86 (Framework) for compatibility.
Faculty adoption	If only one champion exists, the lab is one departure away from cold. Identify and invest in a second internal champion early.
Power and HVAC	A $100K+ tier can draw 4–5 kW peak. Request a room electrical and HVAC review before procurement, not after.
Measurement under-investment	No metrics → no follow-on funding. Propose a metrics dashboard in the next deliverable.
Branding drift	If the lab gets called "Jon's lab" it has a ceiling. Name it for Oakwood from day one.
Equity inside the lab	A 10-seat lab in a department with 100+ majors needs a clear, transparent admission policy. Suggest a first-cohort rubric: mix of class years, gender balance, mix of CS-major and adjacent.

FUNDING

08 Where the next dollars *actually live.*

If Dr. Smith already has internal funding for the pilot, the conversation about *follow-on* dollars is the conversation that secures the lab's second year. Bring these up only if asked, or if the $200K tier is being discussed.

NSF · HBCU Undergraduate Program (HBCU-UP)	Direct support for STEM curriculum and infrastructure at HBCUs. Match the lab to a proposed retention or capstone outcome.
NSF · CISE Education and Workforce	Specifically funds CS curriculum innovation. The 8-module spine fits the rubric.
NSF · EAGER	Smaller, exploratory grants ($150K–$300K). Good for the pilot+expansion story.
NVIDIA Academic Hardware Grant Program	Hardware credit for academic use. Apply once the pilot is live.
Apple Education	Volume discount, not grant. Worth the call.
Microsoft AI Skills Initiative	Curriculum partnership for AI in higher-ed; worth pairing with Azure credits.
Huntsville workforce ecosystem	NASA Marshall, Redstone Arsenal contractors, biotech corridor — defense/aerospace and bio sectors are actively hiring AI-capable graduates. Internship pipelines are a Year-1 deliverable.
Adventist Health / faith-network philanthropy	For projects that intersect health/community/faith. Sabbath Echo overlap is genuine here.
Private foundations	Schmidt Futures, Patrick J. McGovern Foundation, Mellon (HBCU-specific), Lilly Endowment. Pilot deliverables become the application.

NEXT

09 What to offer *before you leave.*

End the meeting with a *specific* next deliverable, with a *date*. Generic follow-ups are how warm meetings cool. Pick one or two from this menu based on what she signals.

Detailed procurement memo	Line-item vendor quotes for the chosen tier, including education pricing paths and shipping/install line. ~3 days.
Scaled room layout	Floor plan to actual dimensions with power, network, AV, furniture, and zone-by-zone occupancy plan. ~5 days.
8-week pilot curriculum	Weekly module outline, deliverable per week, assessment rubric, faculty time estimate per session, and student application form draft. ~7 days.
Governance policy templates	Lab use policy, data classification taxonomy, prompt-injection incident playbook, FERPA-adjacent data handling rules, and audit log spec. ~5 days.
Success metrics dashboard	KPI definitions, instrumentation plan, semester-1 baseline targets, and reporting template. ~3 days.
Grant narrative draft	A 2-pager that adapts the deck for NSF, NVIDIA, or a foundation submission. ~5 days post-pilot launch.
Industry partnership intro	Two warm intros into the Huntsville aerospace/defense or biotech corridor with a student-internship hook. ~10 days.
Student recruitment plan	Cohort 1 application, rubric, communications plan, faculty referral process, and information-session script. ~5 days.

A final note to yourself

Oakwood students are not behind. They're underserved by the speed at which AI shifted the ground under everyone. A 10-seat lab is small enough to land in this conversation and large enough to change the trajectory of every student who walks through it. You're not pitching hardware. You're proposing a room where Oakwood students do work they'll point *to for the rest of their careers. That framing wins.*

Go in warm. Listen first. Translate every spec into a student outcome. Close with a date.